Hearing Drones in 3D: Convolutional Detection and Direction-Finding from a Reduced Microphone Array

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12 October 2026

Project Category: Applied: Audio / Convolutional Networks / Defense & Security.

Code: github.com/danmasad/Acoustic-Drone-Detection

1. Motivation and problem definition

Drones are small, quiet targets that often fly without a radio link, which defeats cameras in poor visibility and radio detectors outright. Sound is a cheap fallback: rotors are loud and harmonically distinctive. From a ground array we want a report, ten times a second, of whether a drone is audible and in which direction. Published systems use 24 or more synchronized microphones; six in one off-the-shelf recorder would cost about four times less. Rodríguez et al. (2026) name temporal context as untested future work, so the question was whether a few frames of history could let six microphones match twenty-four.

They can, and the history is not why. Six microphones on one recorder match the full array on every drone frame. Extra frames help only where microphones are scarce. A kinematic filter after the network removes lobe hops. It does not produce the far-range accuracy the paper reports, and it does not fix the near field.

2. Related work

Classical localization steers the array and reads the strongest direction. Our baseline is **delay-and-sum**: undo the delays a candidate direction implies, sum, and take the energy. A correct guess adds in phase. The map over azimuth and elevation has a second failure mode, a **sidelobe**, whose delays line up almost as well as the true source. One frame cannot tell the phantom from the source, and a thinner array makes more of them.

The reference paper treats that map as an image and localization as segmentation: a U-Net predicts a mask, and the mask's centroid is the estimate. We adopt that because the array is then a preprocessing choice. The paper uses one frame, scores detection only by whether the direction sits still, and released data and beamforming scripts but no training code. Everything here is reimplemented from the description.

3. Data

The Western Sydney University UAV Acoustic Localization Dataset (CC BY-SA 4.0): a DJI Air 3 recorded on a 24-microphone array, four recorders of six, 48 kHz, with GPS logs at 10 Hz. Splits follow the paper. Training is 52 minutes at site 1. Test 2 is 20 minutes at the same site. Test 1 is 3 minutes at another site and was sealed through development. Row k of a log is frame k of the segment. Targets are 10° discs.

The release labels about five minutes of no-drone audio. We mined pre-takeoff audio the official splits never use, checked by beamformed contrast that those windows are ambient, and dropped 20 s for motor spin-up. Training negatives went from 4.8 to 8.2 minutes and from one scene to three. That is still not enough to learn a detector.

4. Method

One pass caches every frame's spectra (381 bins, 200–4000 Hz). Frequency-domain delay-and-sum then builds maps of 23 elevations by 91 azimuths at 4°, averaged into 16 sub-bands over 200–2200 Hz. The array is unambiguous only below 1.47 kHz, so the maps run past the alias-free band, and even all 24 microphones leave a 3.6 dB phantom beside the true peak.

The model is a depth-3 U-Net in PyTorch, with 3×3 convolutions, circular padding on azimuth, a per-pixel mask, and a pooled presence head. About 2.0 million parameters. Temporal context is K frames stacked on the channel axis, newest last, so it is causal. Hyperparameters follow the paper where we could, and they are frozen for every cell: Adam at 5×10^{-3}, no schedule and no weight decay, batch 32, batch normalization, dropout 0.1, class-weighted cross-entropy. About 1.4% of pixels are the drone, so an unweighted loss is minimized by predicting empty sky. Early stopping watches the test-2 hit rate, with patience 10, at least 15 epochs and at most 60, and restores the best checkpoint rather than the last. There is no separate validation split: test 2 is both the selection set and the main reported flight. Test 1 is not read until the other choices are fixed. The baseline is brightest-pixel delay-and-sum on

the same frames. A separate 1-D CNN (86k parameters) detects from one microphone, so it can be tested on audio we did not render.

4.1 What we kept, and the order we decided it

We started from the paper's network so that a gap would be the array or the time. Their U-Net reads the beamformed map as an image, and localization is a segmentation. The array is then which microphones go in, and six and twenty-four share one architecture. Depth, Adam, and the learning rate stay frozen across every cell.

Two changes were measured before the grid, and the grid uses both. 32 base filters beat their 64, 72% against 63% within 10° on that check, and trained faster, so the narrower model is the one we report. A plain ReLU in the presence head died in one seed: localization still reached 72% within 10°, and detection sat at chance from the first epoch. The head is a leaky ReLU with slope 0.01, and the grid was restarted so every cell shares it.

History is K frames stacked as channels. A recurrent network was the alternative we did not take. The label is the mask of the current frame, so the present and a few tenths of a second ago should not share a weight, and a window under a second does not need unbounded memory. Stacking gives each slot its own kernel, and the stack is causal because it contains only the present and the past. We trained K of 1, then 5, then 9. Five frames first read as a null, inside the seed noise. Deleting the past of weights that were already trained separated the two effects: a wider input costs accuracy before any history is used, and the history earns that cost back, mostly past 60 m. Nine frames, 0.9 s, is where that term wins, and only on the small array. Shuffling or reversing the frames afterwards costs about 0.2 points, so the cue the network actually uses is persistence. The true peak sits still on the grid while a sidelobe flickers.

We stop on the hit rate, not on the loss. Training loss is still falling at the epoch we keep, and the loss on test 2 is rising, because the loss is a weighted sum over every pixel and a few confident mistakes dominate it. The decoded direction does not fall with that loss. Polar discs, in place of the rectangular map, and a Tversky loss, in place of weighted cross-entropy, were trained. Neither moved the claim; the grids are in notes 32, 36, and 47. A six-microphone subset chosen by a noise-free sidelobe search over 106,183 candidates then trained at 52%. The six we report is device 3, one physical recorder, because that is an array a deployment can field. The kinematic filter is applied to the decoded direction after training. It is not another model.

The kinematic filter runs after the mask, on the decoded direction. It is not an input channel and not a reason to retrain. A DJI Air 3 in sport mode is published at 21 m/s, so a frame of 0.1 s allows 2.1 m of travel. At the GPS range that is a sky-angle cap, plus one 4° cell because the decoder sits on the grid. The stored direction updates at once when the new peak lies inside the cap, and only after five consecutive frames (half a second) when it does not. A hard hold with no second chance was tried; it wrecks the score, because some large jumps are the model correcting itself. A fixed 16° cap, the same rule with no range, landed within about a point of the range-dependent cap on the same-site 24-microphone run.

5. Experiments and results

The error on a frame is the angle between the predicted and true direction, each written as a unit vector. Azimuth and elevation are not scored apart: a compass flip overhead can be a small step on the sky. **Two summaries of that angle do not convert, and this report never puts them in one table.**

- **Against the paper.** Their summary only. The mean angle, their bins 0–50 / 50–100 / 100–200 m, and, on U-Net rows, their stability cut: the last three directions lie within 10°. Their delay-and-sum bars are every frame. Our row in that table uses the same cut.
- **Against our own variants.** Every drone frame, not the stable subset. The share whose sky angle is at most 10°, our bins 0–30 / 30–60 / 60–90 / 90–120 / 120–180 m, and the seed spread. Alongside that, the sideways miss range × sin(angle).

A mean is pulled by the tail. On the same-site flight, delay-and-sum past 100 m has a mean of 31°, next to their 33°, and 2% of those frames are inside 10°. Quoting one number as the other flatters whoever picked it. The new-site filter and the new-site paper bins below are a **second look** at dumps that had already been scored, not a new sealed result. "±" is the seed standard deviation, omitted when it rounds below 2. The pooled within-cell seed spread on the matched site is ±1.2 percentage points; a smaller gap is not a result. Three seeds cannot return a permutation p below 0.10, so gaps across separately trained models are point estimates. The one comparison that survives block resampling is named below.

5.1 Against the paper, on their metric

Delay-and-sum on all 24 microphones reproduces their bars within a few degrees, including the non-monotonic far bin, and their stability rule calls 67.0% of no-drone frames detections against our 68.6%. That check exercises geometry, recorder offsets, the STFT, the beamformer, the labels, and the metric against an independent implementation.

Mean angular error in degrees. U-Net rows are stable frames; delay-and-sum rows are every frame. The new-site block is a second look. Our matched-site U-Net range is three seeds (notes 23).

	Site	0–50 m	50–100 m	100–200 m
Paper, delay-and-sum	matched	38°	75°	33°
Ours, delay-and-sum, 24	matched	40°	72°	31°
Paper, U-Net, stable	matched	10°	13°	14°
Ours, U-Net 24, stable	matched	7–10°	9–12°	31–74°
Paper, delay-and-sum	new, second look	8°	29°	92°
Ours, delay-and-sum, 24	new, second look	9°	23°	83°
Paper, U-Net, stable	new, second look	12°	24°	54°
Ours, 24 mics K=1, stable	new, second look	16±4°	10±3°	20±9°
Ours, device 3 K=1, stable	new, second look	14°	12±3°	11±2°

Where that leaves the comparison. On the matched site, under 100 m, we are level with them and slightly ahead on the frames they keep (7–10° and 9–12° against 10° and 13°). Past 100 m there, their 14° still beats our stable-frame mean (31–74°). The filter does not change that sentence: it is not their cut, and on their metric the far bin on the matched site remains theirs. On the new site, second look, they lead under 50 m and we lead from 50 m out. Device 3 at one frame is 12±3° and 11±2° against their 24° and 54° on the frames they keep. That flight is 1,592 drone frames, about 2.7 minutes, with 238 frames past 100 m. It is a direction, not a second sealed trial.

5.2 Against ourselves, on every frame

Share of drone frames inside 10°, matched site, our bins. Each entry is the raw decode, then the same frames after the filter. The remaining arrays, and the polar and Tversky grids, are in notes 47.

	0–30 m	30–60 m	60–90 m	90–120 m	120–180 m	All
Delay-and-sum, 24	50→52	33→36	18→21	3→0	2→0	33→35
24 mics, K=1	66±5→66±5	79→83±3	73→82±6	57±3→74±12	27±11→55±13	70±2→76
24 mics, K=5	63→62	79±3→81±2	75→77±3	60±2→64±6	27±6→34±12	70→72
Device 3, K=1	69±3→70±2	75→81±2	68→77	56→64±20	25±4→65±11	69→75±2
Device 3, K=9	72±2→72±2	78±3→78±4	73±3→74±4	71±12→74±12	24±6→22±4	73→73±2

Sideways miss for delay-and-sum on the matched site, our reading only. A 10° error is about 5 m at 30 m and 17 m at 100 m. The far-bin entry is that bin's 32° mean. Not a comparison with the paper.

	Under 30 m	Far bin
Typical sideways miss, 24 microphones	3 m	64 m

Six microphones match twenty-four. At one frame the gap is 69.3% ± 0.7 against 70.5% ± 1.5, inside seed noise;

at five frames it is still inside that noise. At nine frames, 0.9 s, device 3 reaches 73.2% ± 1.4 and all 24 stays at 70.1% ± 2.7. Depth pays only on the small array. The six a noise-free search called optimal scores 52.1% ± 0.8 once trained. The search used no audio: 106,183 non-coplanar subsets, a plane wave in 250 directions, scored by the strongest response more than 15° off the source. The proxy was over-fit: it predicts the arrays it did not select at −0.80, and including its own winner collapses that to −0.36 (notes 22). Device 3 was not that winner. It is one recorder.

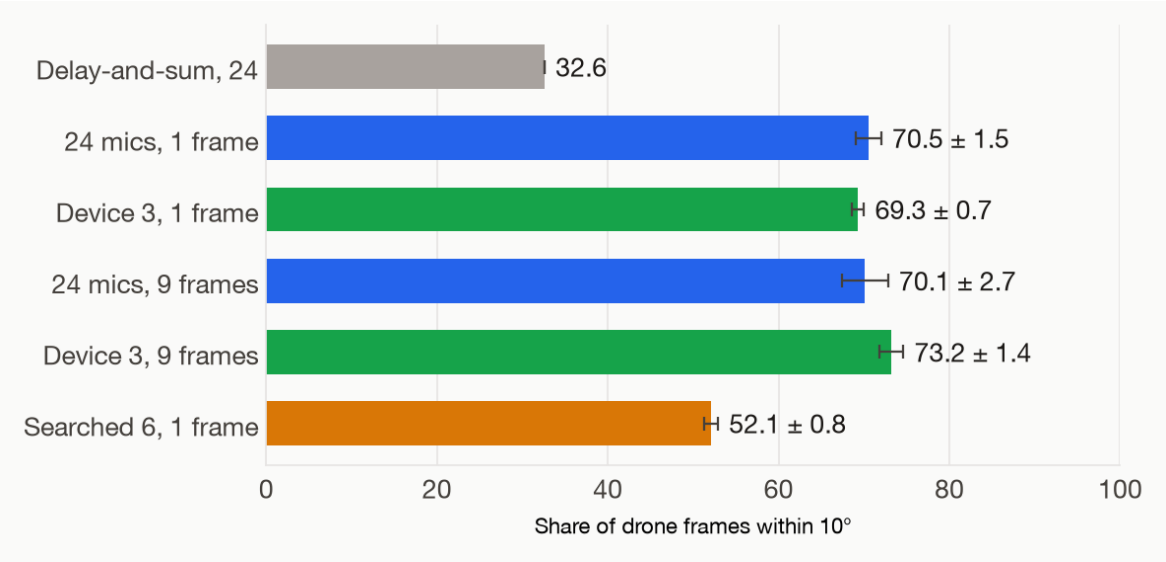


Figure 1: Share of drone frames within 10° on the matched site, before the filter. The bar is the standard deviation across three seeds. Delay-and-sum is one run.

The filter, on this reading. Same site, every frame: 24 microphones at one frame go from 70% to 76% inside 10°, and device 3 at one frame from 69% to 75%. Once five or nine frames are already stacked it adds about a point, and device 3 at K=9 stays at 73%. The gain is in the far bins. Inside 30 m the cells do not move. On the new site, second look, the same one-frame filter goes from 52% to 61% (24 microphones) and from 54% to 61% (device 3), and device 3 at K=9 stays at 52%. The single-frame network plus the filter is the combination that moves; the stack learned the same smoothness on the training site and did not carry it. Where the raw track is stably wrong, the filter holds the error. It is not a repair for a bad array (notes 47).

`utils/analysis/score_kinematic_filter.py` regenerates the filter and the sideways-miss numbers.

History measured on weights that are already trained agrees with the table. Corrupting the past of the K=5 models costs 4.3 percentage points, in shrinking steps, and the climb ends near 0.8 s. Deleting history from the K=9 weights costs 5.1 points. The test is paired on one set of weights, and neighbouring frames are resampled in blocks because they are not independent trials. Holm-adjusted $p = 0.0007$: about +0.5 inside 30 m, +8.7 at 60–90 m, and +21.8 at 90–120 m. Order does not matter. The cue is persistence. Lobe hops are what flicker, and they are what the filter removes. The smooth miss inside 30 m is the near field, and the filter does not fix it.

Detection fails across scenes. Held out one scene at a time, the per-channel detector scores 0.488 ± 0.056 AUROC. The 0.662 measured earlier was the recording: training and test ambient were two halves of one afternoon. Precision–recall does not rescue this. Drone frames are about 78% of the matched site, so a model that says "drone" on almost every frame has precision near 0.8. At the threshold that keeps 95% of drones, the map models fire on 84–96% of empty frames and precision is 0.78–0.80, the base rate (notes 43). The cross-scene AUROC is the number that does not hide that. Time and extra ambient both sharpen the training site and do not travel; the false-alarm wall on other soundscapes is in notes 30 and 38–39. Past about 90 m the drone is often not the loudest peak in the map, so neither detector nor localizer has a lock, and two thirds of labelled positives there are quieter than typical ambient. On the sealed new site the same one-microphone detector, averaged over all 24 channels, reads 0.639. One channel is 0.53. The quiet audio there is one minute of one scene, so this repeats the within-scene result at a smaller sample.

6. Analysis

Uniform delay-and-sum weights a badly placed microphone as heavily as a good one. All 24 has a coplanar recorder and a pair 0.17 m apart, so the map repeats over much of the band. After the network learns to discount that, eighteen further microphones buy nothing measurable, and the baseline has the same ordering. The 3.6 dB phantom is the ceiling those extra channels do not remove.

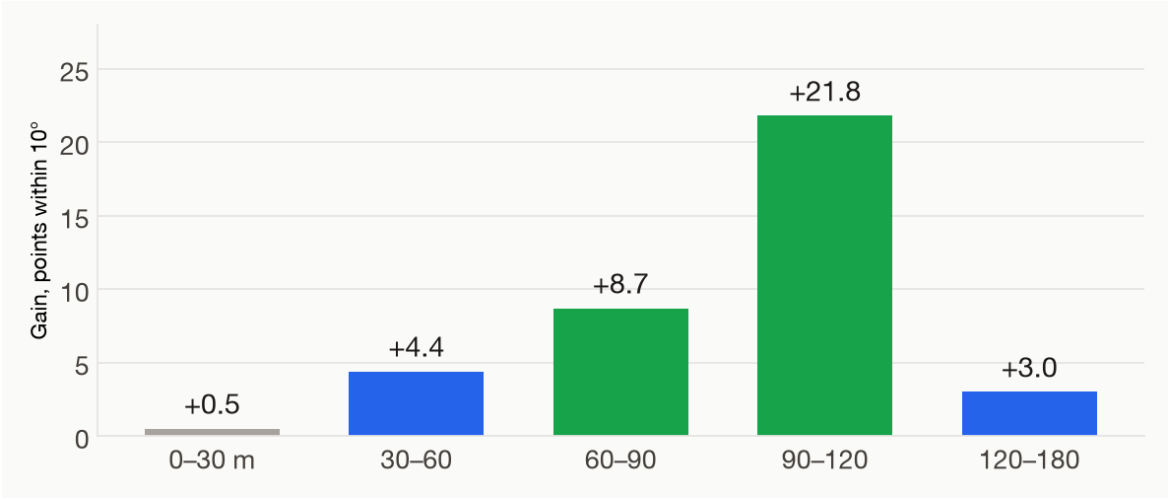


Figure 2: Gain from giving the same trained weights more real past frames, device 3 at nine frames, matched site. The weights and the input width stay fixed. The pooled gain is 5.1 points.

Comparing two trained models confounds effects of opposite sign. Widening the input costs accuracy before any history is used, and history earns that cost back, mostly past 60 m. Section 5.2's small K=5 gap is that net.

Localization is supervised by geometry: every drone frame carries a direction. Detection is supervised by scenes, and we have three. More frames of one field are near-duplicates. An 86k-parameter network at chance across sites, beside a U-Net that localizes, is a data fact. The fix is another afternoon in another place.

7. Limitations

- **Past about 90 m** the labelled drone is often quieter than ambient in the beamformed map. Early stopping on a pooled hit rate never selected for that bin. On the matched site, their stable-frame 14° remains ahead of us there.
- **Inside 30 m** the learned cells sit below the next bin, and the filter leaves them there. On the new site, second look, the closest 30 m stay near 30% inside 10°. Steering assumes a plane wave, which a source metres from a 2 m array is not. Untested.
- **Polar discs, Tversky loss, and the rest of the array list** were measured. None changes the claim above; the grids are in notes 32, 36, and 47. Sub-band count moves the hit rate by about half a point.
- **One drone, one dataset, two sites.** The sealed new-site score, once, already said device 3 is not worse than all 24 (54.4% ± 4.6 against 52.2% ± 5.5 at one frame) while the seed spread widens to ±5–8 points and delay-and-sum wins the pool, because most of that flight is inside 90 m. The paper bins and the filter on that flight, above, are a second look at those dumps. A third site, and ambient from more than three scenes, are what would move the two results that did not transfer: the near field, and detection.

8. Contributions and reproduction

Dan Masad framed the cheap-array question, wrote the pipeline the tables run on (dataset, cache, delay-and-sum, U-Net, detector, metrics), ran the experiments those tables quote, including the one sealed test-1 score, and wrote the original sections 1–7. **Shai Livshits** wrote the four-phase review and notes 43–47: the no-retrain error analyses and the kinematic filter with the two readings in section 5. The new-site rows marked as a second look reuse the dumps from that sealed score. Sealed tables regenerate from the `GATE PASS` scripts in `README.md`. The filter and the sideways-miss numbers regenerate from `utils/analysis/score_kinematic_filter.py`. `docs/design_notes.md` is the lab record.

9. References

B. J. Rodríguez et al., *Acoustic Imaging for UAV Detection*, arXiv:2508.00307, IWAENC 2026, with its dataset. Ronneberger et al., *U-Net*, MICCAI 2015. Thiemann et al., *DEMAND*, 2013. Piczak, *ESC-50*, 2015.